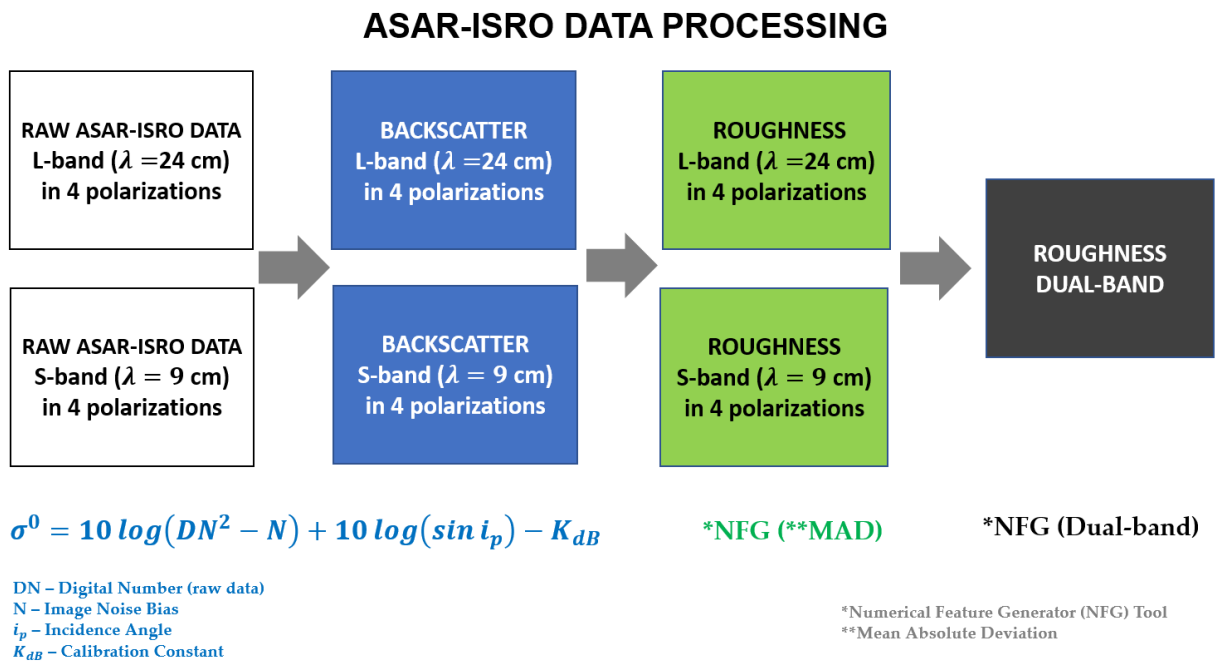


Table 1 IMAGE NOISE BIAS VALUES

STRIP ID	BAND	IMAGE NOISE BIAS (N) AT DIFFERENT POLARIZATION MODES			
		HH	HV	VH	VV
27124	L	1.762430	1.232394	1.891561	1.325989
	S	28.939447	18.862608	22.566810	26.170986
27126	L	1.324492	1.353610	1.421536	1.456411
	S	27.497353	19.322093	21.447556	26.815105
27127	L	0.655546	0.645208	0.703577	0.694209
	S	14.859436	11.114200	11.399114	15.169984
27128	L	0.6611792	0.6476600	0.710281	0.696782
	S	16.934081	12.436480	12.830074	16.764993

*Calibration Constant used for all polarizations KdB = 42.0000

STEP 1 - BSK: Convert amplitude to Backscatter for the entire strip

- Load the entire ASAR strip e.g., for ASAR_ INPUT_PATH=**"/mnt/c/Users/kikou/Desktop/new_SA_Iwp/raw_strips/L_band/124/L_RAW/ASAR_L_JOINT_FP_ID27124_LINE04_RUN01_161219_LEVEL2_HH_V1.3.tif"** *%.tif containing amplitude data, or a directory****
- Load correct txt. GRID file for the same ASAR strip e.g., ANGLE_FILE=**"/mnt/c/Users/kikou/Desktop/Analyses/2_INCIDENCE_angle/Incidence_Angles_all_strips/ASAR_L_S_JOINT_FP_ID27124_LINE04_RUN01_161219_LEVEL2_GRID_V1.3.txt"** *%.txt containing lat,lon,slant angle,incidence angle*

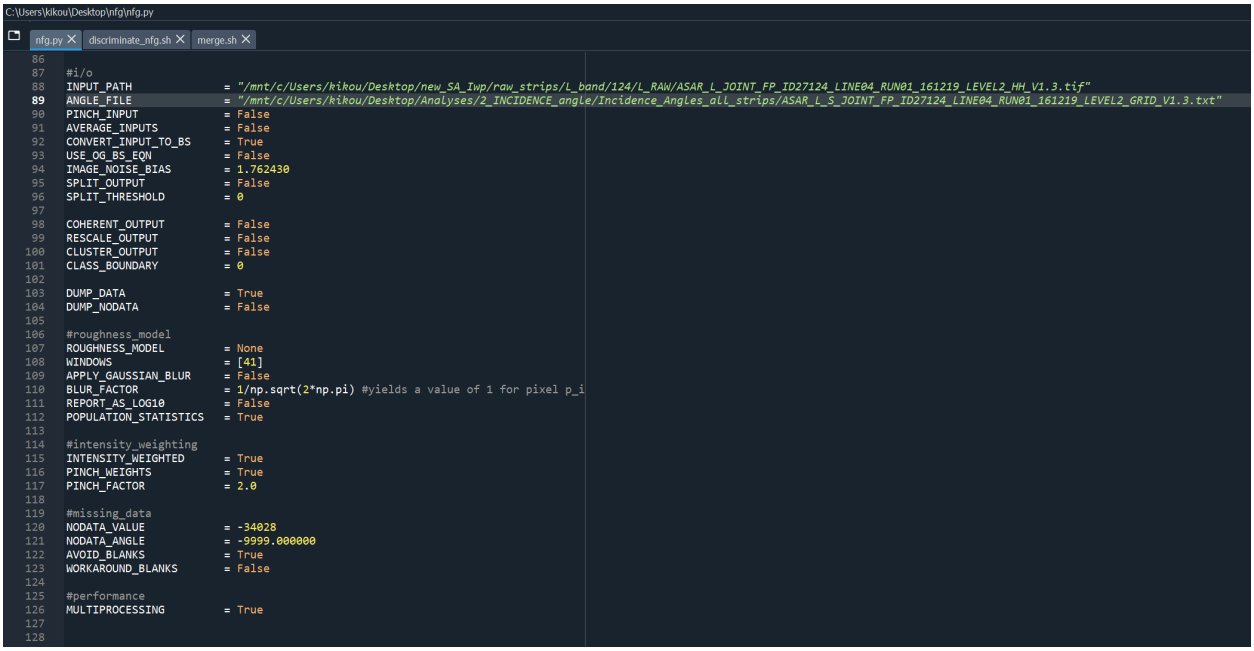
- PINCH_INPUT **= False** *%clip the input data using the pinch factor*
- AVERAGE_INPUTS = **False** *%given a directory of input, yield a single output*
- CONVERT_INPUT_TO_BS = **True** *%toggle to convert amplitudes to backscatter*
- USE_OG_BS_EQN = **False** *%use the old equation provided by NASA to convert to BS: 10*log10(DN^2)-Kdb)*
- IMAGE_NOISE_BIAS = **1.762430** (i.e., correct for ASAR_27124_L_HH, see Table 1) *%the noise in the raw data to be used for BS conversion*
- SPLIT_OUTPUT = **False** (used in Dual-band step) *%produce two outputs contain values opposite of the threshold*
- SPLIT_TRESHOLD = **0** (used in Dual-band step) *%the value to use for spliting output into two*
- COHERENT_OUTPUT = **False** *%treat directory of inputs as a single input for rescaling**
- RESCALE_OUTPUT = **False** *%rescale final output to the range [0,1]*
- CLUSTER_OUTPUT = **False** *%based on the class boundary perform binary classificaion*
- CLASS_BOUNDARY = **0** *%the boundary between smooth and rough used for clustering*
- DUMP_DATA = **True** *%create a .dat column file containing all output pixel values*
- DUMP_NODATA = **False** *%include NODATA_VALUE pixels when dumping data*
- ROUGHNESS_MODEL = **None** or select model number 1-9 but not during the Backscatter step (STEP 1) ONLY use MODELS it for STEP 3 *%the index of the roughness model for the run*
- WINDOWS = [3] Window size is user defined e.g., 3, 5, 7, 9, 15, 21, 27, 41… remember that the smallest Window number is 3 in order to apply the model e.g., standard deviation or mean absolute deviation *%the dimensions for the NxN window** used for computing roughness*
- APPLY_GAUSSIAN_BLUR = **False**
- BLUR_FACTOR = 1/np.sqrt(2*np.pi) *%yields a value of 1 for pixel p_i (it will only work if True above)*
- REPORT_AS_LOG10 = **False** *%toggle for reporting roughness as log10(roughness)*
- POPULATION_STATISTICS = **True** (In our project we use Population rather than Sample Statistics, so it should always be 'True') *%toggle between N & N-1 denominators for stddev etc.*
- INTENSITY_WEIGHTED = **True** (It should be selected as True only for STEP1 Backscatter, as it can apply Intensity of Amplitude during the Backscatter computation) *%weight final output using inverted backscatter values*
- PINCH_WEIGHTS = **True** (gives better histograms, as it adjusts data 'tails') *%apply the pinch factor to constrain intensity weights*
- PINCH_FACTOR = **2.0** *%the number of standard deviations to constrain data to*
- NO_DATA_VALUE = **-34028** (only for ASAR-ISRO and should be changed for other sensors) *%the NODATA_VALUE used in the input and the output*
- NODATA_ANGLE = **-999.00000** *%the magic number used in the angle file indicating nodata*
- AVOID_BLANKS = **True** *%do not compute roughness anytime the window is missing data*
- WORKAROUND_BLANKS = **False** *%recursively reduce window size to remove missing data*
- MULTIPROCESSING = **True** *%toggle for enabling multiprocessing*

*also applies to intensity weighting

**multiple windows can be specified and the final output will be the average value over all windows for a given pixel

***if a directory is specifed, the output will be an average computed over each tif contained in the directory if AVERAGE_INPUTS is True, otherwise each tif contained therein will be treated independently

Repeat this step for all polarizations (HH, HV, VV) and remember to change the Image Noise Bias value, as it varies for all polarisations and wavelengths (see Table 1 above)



STEP 2 – M9: Fill-in the Backscatter blanks

Because the Image Noise Bias values were applied, as well as Incidence Angle correction, some pixels may turn blank. To correct this, we run MODEL 9 at Window [3] to fill-in the blanks by 'smoothing' the Image using nearest neighbors.

Instead of applying this on the entire strip, you can crop the strip to a STUDY AREA of interest and the procedure would be much faster as you are working with much smaller area than the entire strip. Nonetheless, if you prefer, you can apply this step on the entire strip but be aware that it will take long time to complete the step.

- Load the ASAR clip e.g., SA_01
  INPUT_PATH=**"/mnt/c/Users/… "**
- There's no need for Angle File anymore so select 'None'. Please note that Angle File can be only applied to the entire strip as those two files are of same dimensions. You cannot apply Angle file on a smaller clipped area of the strip.
  ANGLE_FILE **= None**
- PINCH_INPUT **= False**
- AVERAGE_INPUTS = **False**
- CONVERT_INPUT_TO_BS = **False**

- USE_OG_BS_EQN = **False**
- IMAGE_NOISE_BIAS = **0**
- SPLIT_OUTPUT = **False**
- SPLIT_TRESHOLD = **0**
- COHERENT_OUTPUT = **False**
- RESCALE_OUTPUT = **False**
- CLUSTER_OUTPUT = **False**
- CLASS_BOUNDARY = **0**
- DUMP_DATA = **True**
- DUMP_NODATA = **False**
- ROUGHNESS_MODEL = **9**
- WINDOWS = **[3]**
- APPLY_GAUSSIAN_BLUR = **False**
- BLUR_FACTOR = 1/np.sqrt(2*np.pi) #yields a value of 1 for pixel p_i (it will only work if True above)
- REPORT_AS_LOG10 = **False**
- POPULATION_STATISTICS = **True**
- INTENSITY_WEIGHTED = **False**
- PINCH_WEIGHTS = **False**
- PINCH_FACTOR = **2.0**
- NO_DATA_VALUE = **-34028**
- NODATA_ANGLE = **-999.00000**
- AVOID_BLANKS = **True**
- WORKAROUND_BLANS = **False**
- MULTIPROCESSING = **True**

STEP 3 – M7: Convert Backscatter data into ROUGHNESS

- Load the ASAR clip e.g., BSK_M9
  INPUT_PATH**="/mnt/c/Users/… "**
- There's no need for Angle File anymore so select 'None'.
  ANGLE_FILE **= None**
- PINCH_INPUT **= False**
- AVERAGE_INPUTS = **False**
- CONVERT_INPUT_TO_BS = **False**
- USE_OG_BS_EQN = **False**
- IMAGE_NOISE_BIAS = **0**
- SPLIT_OUTPUT = **False**
- SPLIT_TRESHOLD = **0**
- COHERENT_OUTPUT = **False**
- RESCALE_OUTPUT = **True** (for roughness models, always rescale the output i.e., 0-1)
- CLUSTER_OUTPUT = **False**

- CLASS_BOUNDARY = **0**
- DUMP_DATA = **True**
- DUMP_NODATA = **False**
- ROUGHNESS_MODEL = **7**
- WINDOWS = [15] Window size is user defined e.g., 3, 9, 15, 21, 27, 41… remember that the smallest Window number is 3 in order to apply the model e.g., standard deviation or mean absolute deviation
- APPLY_GAUSSIAN_BLUR = **False**
- BLUR_FACTOR = 1/np.sqrt(2*np.pi) #yields a value of 1 for pixel p_i (it will only work if True above)
- REPORT_AS_LOG10 = **False**
- POPULATION_STATISTICS = **True**
- INTENSITY_WEIGHTED = **False**
- PINCH_WEIGHTS = **True**
- PINCH_FACTOR = **2.0**
- NO_DATA_VALUE = **-34028**
- NODATA_ANGLE = **-999.00000**
- AVOID_BLANKS = **True**
- WORKAROUND_BLANS = **False**
- MULTIPROCESSING = **True**

STEP 4 – Upper and Lower limits maps - repeat this step for L and for S band rougness data of the same target, as this mode will save two maps for L-band (upper and lower threshold limit maps) and the same for S-band (upper and lower limit maps). Here, we used SPLIT_OUTPUT = True and SPLIT_THERSHOLD = 0.5. After this proceed to STEP 5 to merge upper and lower results into one dual-band map.

- Load the L or S-band roughness clip
  INPUT_PATH**="/mnt/c/Users/… "**
- There's no need for Angle File anymore so select 'None'.
  ANGLE_FILE **= None**
- PINCH_INPUT **= False**
- AVERAGE_INPUTS = **False**
- CONVERT_INPUT_TO_BS = **False**
- USE_OG_BS_EQN = **False**
- IMAGE_NOISE_BIAS = **0**
- SPLIT_OUTPUT = **True**
- SPLIT_TRESHOLD = **0.5**
- COHERENT_OUTPUT = **False**
- RESCALE_OUTPUT = **True** (for roughness models, always rescale the output i.e., 0-1)
- CLUSTER_OUTPUT = **False**
- CLASS_BOUNDARY = **0**

- DUMP_DATA = **True**
- DUMP_NODATA = **False**
- ROUGHNESS_MODEL = **None**
- WINDOWS = [15] Window size is user defined e.g., 3, 9, 15, 21, 27, 41… remember that the smallest Window number is 3 in order to apply the model e.g., standard deviation or mean absolute deviation
- APPLY_GAUSSIAN_BLUR = **False**
- BLUR_FACTOR = 1/np.sqrt(2*np.pi) #yields a value of 1 for pixel p_i (it will only work if True above)
- REPORT_AS_LOG10 = **False**
- POPULATION_STATISTICS = **True**
- INTENSITY_WEIGHTED = **False**
- PINCH_WEIGHTS = **True**
- PINCH_FACTOR = **2.0**
- NO_DATA_VALUE = **-34028**
- NODATA_ANGLE = **-999.00000**
- AVOID_BLANKS = **True**
- WORKAROUND_BLANS = **False**
- MULTIPROCESSING = **True**

STEP 5 – CREATE DUAL-BAND MAP by merging L and S maps (using merge code). For example upper L-band map (containing roughness data above 0.5 treshold) with the lower S-band map (containing roughness data below 0.5 threshold). This will create a complete dual-band map highlighting roughness features visible in L-band alone (e.g., debris avalanche in MSH_TEST) and wet channel within lahar deposits, visible only in S-band. Sometimes, fine tuning is needed (e.g., L-band 0.5 upper and S-band 0.3 lower) to ensure that map is complete (all pixels filled, no blank pixels).